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Duration	1 min 52 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the entire list
b. In the right sublist
c. In the left sublist ✓
d. At the beginning of the list

Question **2**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Inserting elements into a list
b. Sorting elements in a list
c. Determining whether an element is present in a list ✓
d. Deleting elements from a list

Question **3**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted
b. When the list is small and unsorted ✓
c. When the list is small and sorted
d. When the list is large and unsorted

Question **4**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search backtracks to find duplicate elements
b. The search continues until the end of the list
c. The search stops immediately ✓
d. The search starts over from the beginning

Question **5**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search ends successfully ✓
b. The list is sorted
c. The search continues in the right sublist
d. The search continues in the left sublist

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search requires the list to be sorted.
c. Linear search divides the list into halves.
d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **7**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 500
b. 1000
c. 10 ✓
d. 100

Question **8**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 94 and 99

b. 90 and 99 ✓

c. 90 and 100

d. 89 and 94

Question **9**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Hash search

b. Linear search

c. Binary search ✓

d. Sequential search

Question **10**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By comparing each element sequentially

b. By using a hash function

c. By dividing the list length by two ✓

d. By starting from the first element

Question **11**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Linear search ✓

b. Binary search

c. Jump search

d. Hashing

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It does not work with negative numbers
b. It requires the list to be sorted ✓
c. It is slower than linear search for small lists
d. It can only be applied to large lists

Question **13**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all
b. When the item is somewhere in the middle of the array ✓
c. When the item is the last element in the array
d. When the item is not the array at all

Question **14**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list
b. Binary search can find multiple instances of the target element
c. Binary search works on unsorted lists
d. Binary search has a lower time complexity on large, sorted lists ✓

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Must use a sorted array
b. Binary search algorithm is not efficient when the data elements more than 1500 ✓
c. There must be a mechanism to access middle element directly
d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed